The Jacobian Conjecture after the Three-Dimensional Counterexample: Exact Validation, a General Seed Family, the Geometry of Non-Properness, and Two-Dimensional Equivariant Rigidity

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Abstract

On 2026-07-19, Levent Alpöge announced (crediting the language model Claude Fable 5 and a question posed by Akhil) an explicit polynomial map $F : \mathbb{C}^3 \to \mathbb{C}^3$ with constant Jacobian determinant -2 and a repeated value, refuting the Jacobian conjecture for every $N \geq 3$. We report the results of an exact-arithmetic experimental program built on this event. First, we validate the counterexample independently and compute the full fiber over the announced target: it consists of exactly the three announced points. Second, we reverse-engineer the structure of F: it is a weighted skew-product over the invariants $v = xy$, $t = x^2z$, its Keller property reduces to a two-variable identity, and its fibers are governed by a cubic in a single composite variable. Third, we derive and verify a general constructor: every admissible triple (seed polynomial p, scale k, section q) yields a Keller map with determinant $-k\,p(1)^2$ and generic fiber degree $\deg p + 1$; we exhibit new explicit counterexamples with rational collision certificates in fiber degrees 4, 5 and 6, plus a section-tail instance beyond the announced family shape, prove the degree law $(5d-3, 5d-4, 4)$ within the family, and show the announced map is its smallest member. Fourth, we make the failure geometry exact: a preimage escapes to infinity precisely when its fiber root is a multiple root, so the asymptotic variety of F is the plane $C = 0$ together with the explicit discriminant surface $27A^2C^2 - 18ABC + 16A + B^3C - B^2 = 0$; over $\mathbb{R}$ this wall separates a three-sheet from a one-sheet region, and F restricts to a surjective, non-injective real Keller map. The counterexamples reduce at suitable primes to non-injective Keller maps over $\mathbb{F}_\ell$ of degree below the characteristic. Fifth, we prove that the mechanism cannot reach the still-open two-variable case: for arbitrary weights, every $\mathbb{G}_m$-equivariant Keller map of $\mathbb{C}^2$ is linear (a classification with a positive-factor kill at the top degree), so a planar counterexample, if any, must be genuinely non-equivariant. All computations are exact (rational or symbolic) and reproducible from the repository; every claim below is labeled as machine-verified, derived, or conjectural.

1 Introduction

Let k be a field of characteristic 0. A polynomial map $F = (F_1, \ldots, F_N) : k^N \to k^N$ is a *Keller map* if $\det JF$ is a nonzero constant. The Jacobian conjecture $JC(N)$, stated in two variables by

Kraus in 1884 (with a flawed proof whose gap, control of ramification at infinity, remained the subject's central obstacle [6]) and popularized by Keller in 1939 [4], asserted that every Keller map has a polynomial inverse. Over $\mathbb{C}$, invertibility is equivalent to injectivity, so a refutation requires exactly one repeated value. The classical positive ladder includes Wang's degree-2 theorem, Moh's verification of $JC(2)$ up to degree 100, the Bass–Connell–Wright reduction to cubic homogeneous form [2], and Drużkowski's cubic-linear form [3]; Pinchuk [5] refuted the strong real variant by a non-proper local diffeomorphism with non-constant Jacobian. The conjecture is Problem 16 of Smale's list [7].

On 2026-07-19/20, Alpöge announced the map, with $u = 1 + xy$,

$$F = \left(u^3 z + y^2 u(4 + 3xy),\;\; y + 3xu^2 z + 3xy^2(4 + 3xy),\;\; 2x - 3x^2 y - x^3 z \right), \tag{1}$$

with $\det JF \equiv -2$ and $F(0, 0, -\frac{1}{4}) = F(1, -\frac{3}{2}, \frac{13}{2}) = F(-1, \frac{3}{2}, \frac{13}{2}) = (-\frac{1}{4}, 0, 0)$ [1]. Appending identity coordinates refutes $JC(N)$ for all $N \geq 3$; the case $N = 2$ remains open. At the time of writing no formal paper exists; the announcement is a machine-checkable certificate, which makes an exact experimental program both possible and necessary. This manuscript maps the findings of that program (repository: CAOS_RESEARCH, problem `algebraic-geometry/jacobian-conjecture`; experiments cited as EXP-*nnn*). Each claim is labeled: **[MV]** machine-verified exact identity, **[D]** derived analytically and certified on instances, **[C]** conjecture.

2 Exact validation and the full fiber (EXP-001)

Theorem 2.1 ([MV])**.** *For F as in (1): $\det JF = -2$ identically; the three announced points are pairwise distinct with the common image $(-\frac{1}{4}, 0, 0)$; and the fiber $F^{-1}(-\frac{1}{4}, 0, 0)$ consists of exactly those three points. In particular F is a non-injective Keller map and $JC(N)$ is false for every $N \geq 3$.*

The fiber computation is a lex Gröbner basis over $\mathbb{Q}$: the eliminant in z is $\frac{1}{8}(2z - 13)(4z + 1)$, and on the fiber $3x + 2y = 0$, $12y^2 = 4z + 1$. All coefficients of (1) are rational, so non-injectivity holds equally over $\mathbb{R}$.

3 The structure of the counterexample (EXP-002)

Theorem 3.1 ([MV])**.** *F is equivariant for the $\mathbb{G}_m$-action with source weights $(1, -1, -2)$ and target weights $(-2, -1, 1)$. Writing $v = xy$, $t = x^2 z$, $u = 1 + v$, the reduced data*

$$a_1 = x^2 P, \qquad b_1 = xQ, \qquad c_1 = R/x = 2 - 3v - t$$

are polynomials in (v, t); each component of F is affine in z with coefficient vector $(u^3, 3xu^2, -x^3)$; and the generating identity $a_1 = \frac{ub_1}{2} - u^2 + \frac{u^3 c_1}{2}$ holds.

Lemma 3.2 (Skew-product determinant lemma, [MV] with generic symbolic coefficients)**.** *For* any *polynomials a_1, b_1, c_1 in (v, t), the map $(a_1/x^2,\, b_1/x,\, xc_1)|_{v=xy,\, t=x^2 z}$ satisfies*

$$\det JF \cdot c_1^2 = -J_2, \qquad J_2 = \det \frac{\partial(V', T')}{\partial(v, t)}, \quad V' = b_1 c_1,\; T' = a_1 c_1^2.$$

For F, $J_2 = 2c_1^2$ **[MV]**: the Keller property is a *two-variable* identity; the third dimension only carries weights. Fibers are governed by a single cubic: with $w = uc_1/2$ and $\Phi(w) = w^2 - w^3$,

$$\Phi(w) = \tfrac{1}{4}\left(w\,QR - PR^2\right) \quad \text{identically } \textbf{[MV]}, \tag{2}$$

so preimages of (A, B, C) correspond to roots of $4\Phi(w) - wBC + AC^2 = 0$; at the announced target this degenerates to $\Phi(w) = 0$ with roots $\{0, 0, 1\}$, matching the three preimages' w-values $(1, 0, 0)$ **[MV]**. Injectivity dies by *non-properness*: a Keller map is a local diffeomorphism, so sheets merge only at infinity; this is Pinchuk's loophole and the modern face of Kraus's gap.

4 The seed family: a general counterexample constructor (EXP-003/004)

The identification $c_1 = h(v) - t$ with $h = p/w$ is a degree-2 coincidence: an initial constructor built on it is *provably unsolvable* for $\deg p \geq 3$ (EXP-003, an exact unsolvability statement). The failure analysis yields the correct *potential form.*

Theorem 4.1 (Constructor v2, [D], certified on instances [MV])**.** *Let $p(w) = \sum_{i=1}^{d} p_i w^i$ satisfy*

$$p(0) = 0, \qquad \int_0^1 p = 0, \qquad p(1) \neq 0, \qquad p'(1) \neq 2p(1),$$

let $k \neq 0$, $m = -k\,p(1)$, and let $q(v)$ be any polynomial with $q(0) = k$ and $q'(0) = k\,(p(1) - p'(1))/(p'(1) - 2p(1))$, with arbitrary higher-order tail. Set $\Phi(w) = \int_0^w p$, $s = c_1 = q(v) - t$, $w = us/k$, and define

$$V' = k^2 p(w) + ms, \qquad T' = wV' - k^2\Phi(w), \qquad b_1 = V'/s, \qquad a_1 = T'/s^2.$$

Then a_1, b_1 are polynomials in (v, t), and $F_{p,k,q} = (a_1/x^2,\ b_1/x,\ xc_1)$ is a polynomial Keller map of $\mathbb{C}^3$ with

$$\det JF_{p,k,q} = -\frac{m^2}{k} = -k\,p(1)^2 \quad (\textit{constant, for every admissible } (p, k, q)),$$

generic fiber degree $d+1$, and the exact reconstruction inverse: from a root w of the fiber polynomial $k^2\Phi(w) - (w\,BC - AC^2)$,

$$s = \frac{BC - k^2 p(w)}{m}, \quad u = \frac{kw}{s}, \quad x = \frac{C}{s}, \quad v = u - 1, \quad t = q(v) - s, \quad y = \frac{v}{x}, \quad z = \frac{t}{x^2}.$$

*The announced map (1) is the instance $p = 2w - 3w^2$, $k = 2$, q affine; at $d = 2$ this shape is rigid up to the scale k **[MV]**.*

The three admissibility conditions are exactly the low-weight polynomiality constraints at the origin; the potential form makes $J_2 = m^2 s^2/k$ automatic, so *the Keller property costs nothing*: this is the structural reason the conjecture is false in dimension ≥ 3.

Theorem 4.2 (New explicit counterexamples, [MV])**.** *The following maps are non-injective polynomial Keller maps of $\mathbb{C}^3$, each certified by an engineered rational collision:*

Seed p	k	det	*Component degrees*	*Generic fiber degree*
$w - 2w^3$	3	-3	$(12, 11, 4)$	4
$8w - 12w^2 + 4w^3 - 5w^4$	14	-350	$(17, 16, 4)$	5
$2w - 3w^2$, *section tail* v^2	2	-2	$(8, 7, 5)$	3

For the first: $(-\frac{1}{15}, 18, 5130)$ *and* $(\frac{1}{24}, -18, -10368)$ *both map exactly to* $(-54, -54, 1)$*. Collisions are engineered by prescribing two rational roots* $w_1 \neq w_2$ *and solving the linear system* $w_i\tilde{V} - \tilde{T} = k^2\Phi(w_i)$ *for the target invariants.*

The section-tail instance changes the component-degree pattern at fixed fiber degree and determinant: the family is strictly larger than the seed-only family circulating informally. Every negative rational number is realized as a Keller determinant within the family.

Theorem 4.3 (Degree laws and family-internal minimality, [MV] for $d \leq 5$)**.** *For affine sections* q *the component total degrees of* $F_{p,k,q}$ *are exactly* $(5d-3,\ 5d-4,\ 4)$*,* $d = \deg p$ *(verified* $d = 2, \ldots, 5$*; the* $d = 5$ *instance, seed* $w - 3w^5$*,* $k = 5$*,* $\det = -20$*, degrees* $(22, 21, 4)$*, fiber degree 6, carries its own rational collision certificate). The scale* k *never changes degrees; section tails strictly increase them; degree-1 seeds are impossible (the integral condition forces* $p = 0$*), so the fiber-degree floor is 3. Consequently the minimal degree vector over the family is* $(7, 6, 4)$*, attained exactly at the announced map:* the July 2026 counterexample is the smallest instance of its own family*. Globally, only Wang's degree-2 theorem constrains* $\mathbb{C}^3$*; the existence of a counterexample of total degree* 3 *to* 6 *remains open.*

Remark 4.4 (Characteristic p, [MV]). The $d = 3$ instance reduces mod $\ell = 13$ and $\ell = 101$ to explicit non-injective Keller maps over $\mathbb{F}_\ell$ of total degree $12 < \ell$ (the rational collision reduces to distinct points with a common image, e.g. mod 13: $(6, 5, 8)$ and $(6, 8, 6)$ both to $(11, 11, 1)$). Unlike the classical char-p example $x - x^p$ (degree $= p$), the degree sits strictly below the characteristic; expected from standard integer-model reductions, now explicitly certified.

5 The asymptotic variety: where sheets merge

Theorem 5.1 (Escape = multiple fiber root, [MV] on F and the $d = 3$ instance)**.** *In the potential form, the fiber polynomial* $W(w) = k^2\Phi(w) - wBC + AC^2$ *satisfies* $W'(w) = k^2p(w) - BC$ *identically, and the reconstruction scale is* $s = -W'(w)/m$*. Hence a fiber root fails to reconstruct a finite preimage (the preimage escapes to infinity) precisely when it is a* multiple *root of* W*: over* $C \neq 0$ *the non-properness locus is the discriminant locus of* W*. For the announced* F*,*

$$A(F) \;=\; \{C = 0\} \;\cup\; \{\, 27A^2C^2 - 18ABC + 16A + B^3C - B^2 = 0 \,\}.$$

Verified end to end: the target $(0, 1, 1)$ *lies on the surface and its exact fiber is the single point* $(2, -\frac{1}{2}, \frac{9}{8})$ *(two sheets escaped); over generic* $(A, B, 0)$ *the fiber is the single flat-sheet point* $(0, B, A - 4B^2)$*; the collision target* $(-16, -16, 1)$ *lies off the surface with a full 3-point rational fiber. Collisions are generic; escapes are codimension 1. Kraus's "ramification at infinity" is, for this family, the discriminant of a univariate polynomial.*

6 The two-dimensional frontier: obstruction, rigidity, real census (EXP-005/006/010/011)

Theorem 6.1 ([MV] with generic symbolic f, g)**.** *Let $\mathbb{G}_m$ act on $\mathbb{C}^2$ with weights $(1, -a)$, $a \geq 1$, and let $F = (x^{b_1} f(v),\, x^{b_2} g(v))$, $v = x^a y$, be an equivariant polynomial map. Then*

$$\det JF = x^{\,b_1+b_2+a-1} \left(b_1 f g' - b_2 f' g\right),$$

so the Keller condition forces $b_1 + b_2 = 1 - a$ and the single univariate identity $b_1 f g' - b_2 f' g =$ const $\neq 0$.

Proposition 6.2 (One-invariant collapse)**.** *If the reduced data (V', T') of a weighted skew-product depend on a single variable, then $dV' \wedge dT' = 0$, hence $J_2 \equiv 0$ and $\det JF \equiv 0$: no Keller map arises. Since a nontrivial $\mathbb{G}_m$-action on $\mathbb{C}^2$ has one independent weight-0 invariant (against two in $\mathbb{C}^3$), the potential-form mechanism of Theorem 4.1 is structurally unavailable in dimension 2.*

Additionally **[MV]**: no coordinate slice of (1) (each variable frozen at rational test values, any output pair) is a 2D Keller map.

Theorem 6.3 (2D equivariant rigidity, [D], every mechanical step [MV] on sweeps)**.** *For any $\mathbb{G}_m$-action on $\mathbb{C}^2$ with weights $(w_1, -w_2)$, $w_1 \geq 1$, $w_2 \geq 0$, every equivariant polynomial Keller map is linear: $(f_0 x,\, g_0 y)$ or $(f_0 y,\, g_0 x)$.*

Proof shape. Equivariant components are $P = x^{i_1} y^{j_1} f(v)$, $Q = x^{i_2} y^{j_2} g(v)$ with $v = x^{w_2} y^{w_1}$, and the determinant factors as

$$\det JF = x^{i_1+i_2-1}\, y^{j_1+j_2-1} \big[(i_1 j_2 - i_2 j_1)\, fg + v\,(\beta_1 f g' - \beta_2 f' g)\big], \qquad \beta_r = w_1 i_r - w_2 j_r$$

(certified on 648 weight-times-exponent cases with generic symbolic f, g). Constancy forces $i_1 + i_2 = j_1 + j_2 = 1$, and the mixed shapes ($P = xyf$, $Q = g$ and its swap) admit no Keller solution (their bracket is divisible by v; empty solves). The surviving shapes reduce to $fg + v(w_1 f g' + w_2 f' g) = c$ (or its swap), whose coefficient at the top bidegree $N = d_f + d_g$ is $(1 + w_1 d_g + w_2 d_f)\, f_{d_f}\, g_{d_g}$ with a strictly positive factor: the leading coefficients die and by descent f, g are constant (certified: empty constrained solves for all exact bidegrees ≤ 5 over eight weight pairs). □

The independent enumeration EXP-006 (216 polynomial Keller instances over a bounded window, all 648 exactly-computed in-image fibers single-point, every instance linear **[MV]**) is the experimental companion of Theorem 6.3. Consequence: a two-variable counterexample, if one exists, must be genuinely non-equivariant: the symmetry class behind the entire three-dimensional family is completely closed in dimension 2.

Theorem 6.4 (Real fiber census of F, [MV])**.** *Over $\mathbb{R}$, the announced F restricts to a surjective, orientation-reversing (*$\det = -2$*), non-proper, non-injective real Keller map. For real targets off the discriminant wall the number of real preimages is 3 (region $D > 0$, which contains the collision targets) or 1 (region $D < 0$); two independent exact routes (Sturm isolation of the fiber cubic plus reconstruction, versus Gröbner elimination) agree at all samples, and no empty real fiber exists on a 36-target rational grid (an odd-degree argument makes this structural). In particular the constant-Jacobian* real *statement in dimension 3 falls with the same example; Pinchuk's planar counterexample (1994) concerned the non-constant case.*

The honest map of remaining routes to a hypothetical 2D counterexample: no symmetry at all; non-reductive ($\mathbb{G}_a$) structures, which face the same one-invariant collapse in naive form; or gradings at infinity in the Newton-polygon/Abhyankar sense. Conversely, a proof of $JC(2)$ in this program's terms is a properness statement: the reconstruction inverse of Theorem 4.1 shows exactly how asymptotic values are produced in dimension 3 (denominators $BC = k^2 p(w)$), and whether one-invariant geometry leaves room for such escape is, in our view, the sharpest transferable open question.

7 The planar machine: uniform closure, descent, and the primitive stratum

The Keller condition $\det J(P, Q) = 1$ is bilinear in the coefficient vectors of P and Q, hence linear in one side given the other. In the affine gauge (identity linear parts) this yields a machine per degree pair (m, n): eliminate the large side (a lexicographic Groebner elimination produces the consistency ideal in the small side's coefficients), parametrize the consistency variety, complete linearly, and invert explicitly or fiber-test. The $(2, 3)$ consistency ideal is the single equation $(4A_0A_2 - A_1^2)^2 = 0$: the top form of P must be a perfect square, exactly the leading-form degeneracy; the $(2, 4)$ elimination returns the same ideal.

Theorem 7.1 (Uniform closure at min degree ≤ 2; machine-verified at the certified degrees)**.** *Write $\ell = \alpha x + \beta y$, $\beta \neq 0$. In shear coordinates $(u_1, u_2) = (\ell, x)$ the Keller condition for $P = x + \ell^2$ is the linear first-order PDE $2u_1 Q_{u_2} - Q_{u_1} = c$ with characteristic invariant P; its complete polynomial solution space is $Q = \ell/\beta + H(P)$, H arbitrary. Consequently every planar Keller map with $\min(\deg P, \deg Q) \leq 2$ is invertible, with the closed inverse $\ell = \beta(v - H(u))$, $x = u - \ell^2$, $y = (\ell - \alpha x)/\beta$.*

Sufficiency is verified generically; completeness as exact kernel dimensions $\lfloor n/2 \rfloor + 1$ ($n = 3..6$, three rational (α, β) samples); the inverse by exact composition. The closure extends verbatim to any section polynomial: the whole quasi-triangular class $P \sim x + f(\ell)$ is invertible at every degree by the same formula with f in place of ℓ^2. At bidegrees $(3, 3)$ the machine also forces the alignment: for $P = x + A_0x^2 + A_1xy + A_2y^2 + ay^3$ the consistency ideal contains A_0^2 and A_1^6 once a is inverted, so a cube-top cubic P admits completions only in quasi-triangular form.

The classical degree descent is, on this machine, an algorithm: while the leading forms are power-proportional, subtract $c\,P^k$ from Q; when the minimum degree reaches 2, finish with the closed inverse of Theorem 7.1; replay the elementary steps on the value side to assemble the inverse of the original map. The implementation explicitly inverted an entire deterministic library of composed Keller maps (0 to 4 steps each), verified by exact composition. Its failure signature is precise: a *primitive* pair, i.e. leading forms sharing a base form h with $\deg h \geq 2$ and no power relationship. $JC(2)$ is exactly the statement that no Keller pair is primitive.

First contact with that stratum, at the smallest primitive bidegrees $(4, 6)$ ($P_{\text{top}} = ah^2$, $Q_{\text{top}} = bh^3$, h quadratic): on a structured sampled slice ($h = xy$ and the rank sweep $x^2 + \gamma y^2$, 25 samples, $a \neq 0$, including a generic symbolic a), the completion window is empty: no Keller partner of degree ≤ 6 exists at all (the complete linear system is inconsistent in every sample), and one descent step extends the exclusion to degree ≤ 8. Positive controls certify the scan is not vacuous. Verified against the primary sources, this replicates with a different instrument classically covered territory: Magnus proved invertibility for $\gcd = 1$ [**?**], and Appelgate–Onishi [**?**] with Nagata extended this to gcd prime (refined coverage: $\gcd \in \{1, 8\} \cup P \cup 2P$); the plane conjecture is classically equivalent to the divisibility statement, which is exactly the primitive-stratum framing above. The genuinely open gcd values start at 9 and 12 (bidegrees like $(18, 27)$ and $(24, 36)$): that composite-gcd ladder is the frontier. First contact is made there: for $h = x^4y^5$, the machine certifies that $P = x + ah^2$ admits NO Keller partner of degree ≤ 27 for any $ae0$ (a single certificate pairing with gcd $-144\, a^2$ over 403 completion unknowns), with the same emptiness across three structurally different degree-9 bases and a reach probe at $(24, \leq 36)$. Stated honestly: these bidegrees lie inside Moh's degree-100 verified range, so the contribution is the explicit per-pair certificate rather than the truth itself; the beyond-Moh rungs are the standing target.

Theorem 7.2 (The weight-class theorem; machine-verified at every step)**.** *For $u \geq 2$, $v \geq 1$, $a \neq 0$, the polynomial $P = x + a\, x^u y^v$ is not a Keller component: no $Q \in \mathbb{C}[x, y]$, of any degree, satisfies $J(P, Q) \in \mathbb{C}^\times$.*

Proof sketch. Under the weights $(v, 1-u)$ the polynomial P is weighted-homogeneous, so $J(P, \cdot)$ shifts weight by a constant and the Keller equation decouples into independent weight classes. The constant 1 is reachable only from the class of y, spanned by the ray $g_s = x^{ks}y^{ms+1}$ with $k = (u-1)/d$, $m = v/d$, $d = \gcd(u-1, v)$; on that ray the operator is banded, $J(P, g_s) = (ms+1)\, e_s + aB_s\, e_{s+d}$ with B_s a positive integer and $e_t = x^{kt}y^{mt}$. For a polynomial Q the class component is a finite sum $\sum_{s \leq S} c_s g_s$: the e_0 row forces $c_0 \neq 0$, the band recursion keeps the chain nonzero, and the row beyond the last entry demands it vanish: a contradiction for every truncation S. □

This is the closed form of every pure-slice window certificate above (all seven measured pairings are its chain products), and it certifies past every verified range at negligible cost (e.g. $\deg P = 135$, beyond the degree-108 floor of the current literature). The proof is elementary once seen and may be folklore; we have not found it stated. Its scope is exactly the one-monomial perturbations; non-monomial slices have genuine multi-edge Newton polygons, and the multi-edge calculus (edge filtration, conjectured upper-triangularity) is the program's next instrument.

Theorem 7.3 (Lower-weight perturbations never rescue; machine-shadowed, one derived step)**.** *Let $u \geq 2$, $v \geq 1$, $a \neq 0$, and let $R \in \mathbb{C}[x, y]$ be any polynomial whose monomials have degree ≥ 2 and $(v, 1-u)$-weight strictly below v. Then $P = x + a\, x^u y^v + R$ is not a Keller component.*

Proof skeleton. The operator $L_{\text{top}} = J(x + ax^u y^v, \cdot)$ raises weight by $u - 1$, strictly more than any perturbation term, so the top-graded equations are governed by L_{top} alone. Its kernel on the class of weight kv is exactly $\langle (x + ax^u y^v)^k \rangle$ (machine-identified; the naive injectivity claim fails precisely there, a refutation recorded in the experiment log). Absorbing these kernel components of Q as $H(P_{\text{top}})$ turns the perturbation into the sources $J(R, P_{\text{top}}^k)$ in the constant's equation; the identity $J(m, P_{\text{top}}^k) = -kP_{\text{top}}^{k-1} J(P_{\text{top}}, m)$ reduces them to $k = 1$, and the chain functional of Theorem 7.2 annihilates every such source (machine-verified on all danger weights at adequate windows; the closed-form general step is the one remaining derived gap). The constant's equation then reduces to the pure chain, contradicted at every truncation. Certificates over $\mathbb{Q}(a, b, c)$ confirm the conclusion directly: with danger-weight tails present, the obstruction pairing remains a pure power of a. □

This covers dense-Newton-polygon families of unbounded degree, far beyond every verified range, while remaining silent, as it must, on quasi-triangular components (their leading monomials violate $u \geq 2$, $v \geq 1$). The frontier moves to perturbations of weight $\geq v$, where the top edge itself deforms.

Theorem 7.4 (Every x-anchored edge falls; leading form unconditional)**.** *Let $z = x^k y^m$ with $k, m \geq 1$ and $E = x\,\varphi(z)$ with $\deg \varphi = g \geq 1$. Then E is never the leading form of a Keller component of $\mathbb{C}^2$, and $P = E + R$, for any polynomial tail R of strictly lower weight, is never a Keller component, at any degree.*

Proof of the leading-form statement. On the y-class ray $g_s = x^{ks}y^{ms+1}$ the operator acts by multiplication: $J(E, g_s) = \big[(ms+1)\varphi(z) + kz\varphi'(z)\big]z^s$ (verified symbolically in the edge coefficients). The class equation therefore asks for a polynomial f with $mz\varphi f' + (\varphi + kz\varphi')f = 1$. If $\deg f = D$, the coefficient of z^{D+g} on the left is $(mD+kg+1)\,\varphi_g\,c_D$, and every factor is nonzero; so $c_D = 0$, and by induction $f = 0$, contradicting the constant. The with-tails statement follows the kernel-absorption route of Theorem 7.3, with kernels now the powers of the full edge E (machine-identified). □

Machine confirmations: full completion windows are empty for binomial and trinomial edges including perfect-square and perfect-cube φ (factorization does not rescue); the $\mathbb{Q}(a, b)$ certificate for the binomial edge is $60a^3$, free of the lower edge coefficient. Consequently no Keller component of $\mathbb{C}^2$ has a Newton top edge anchored at the linear vertex with interior direction: components must carry y-anchored (quasi-triangular type) tops, and the two-variable conjecture is exactly the statement that nothing else lives there. Classifying the y-anchored completions is the program's queued endgame frame.

8 Discussion and program

Why the conjecture is false. In the equivariant class, being Keller is a two-variable potential identity, satisfiable for free at any fiber degree ≥ 3; nothing in the determinant sees properness. **What is essential.** Two independent weight-0 invariants; the potential form; valuation conditions at the origin. **Consequence cascade (verified from primary sources, EXP-016).** The Mathieu conjecture is false for $SU(3)$ (Mathieu 1997: Mathieu for $SU(N)$ implies $JC(N)$; abelian case Duistermaat–van der Kallen 1998); the Gaussian moments conjecture is false (GMC implies JC, Derksen–van den Essen–Zhao 2017); Zhao's vanishing conjecture is false (equivalent to JC); the Image conjecture is false in some dimension (implies the vanishing conjecture); and the full Dixmier conjecture is false (stable equivalence $JC(2n) \Leftrightarrow$ Dixmier(n), Belov-Kanel–Kontsevich and Tsuchimoto: $JC(4)$ false gives Dixmier(2) false; Dixmier(1) remains open). **Next experiments.** EXP-010 (2D widening: general weight pairs, higher degrees); the global-minimality search in degrees 3 to 6 (GPU-widened ansatz enumeration); the Bass–Connell–Wright push-through of F to explicit cubic form; primary-source verification of the consequence cascade.

The weighted landscape: uniqueness of the mechanism (EXP-012)

Generalizing to source weights $(1, -1, -m)$, the determinant lemma $\det JF = x^{\sigma+m}D$ and the pairing reductions $J_2 = \pm c_1^{r_a+r_b-1}D$ hold generically **[MV]**, and the reduced Jacobian vanishes to order $m-1$ along the section. The survey outcome: at $m = 1$ the reduced pair is itself a planar Keller map, so that class is a *bridge* to $JC(2)$ (93 scanned instances, all injective; a collision there would refute $JC(2)$); at $m = 3$ the class $o = (-2, -2, 1)$ is *empty* (a one-line valuation proof, machine-certified) and $o = (-3, -1, 1)$ is rigid in scans (60 instances); at $m = 4$ the potential-form family does *not* exist: with seed degree up to 5 the admissibility system has Gröbner basis [1] under $\mu \neq 0$ **[MV]**. Our design hypothesis (a parity law producing families at every even m) was thereby refuted by its own experiment; the finding is stronger: *within the scanned weighted landscape, the announced counterexample's weight system is the unique collision-capable mechanism.* Scope: sparse bounded scans, quantified in the experiment record; wider weight systems and higher-degree seeds are queued.

Reproducibility

Every theorem labeled [MV] is an exact sympy computation over $\mathbb{Q}$ with a persisted script and output in the repository (experiments EXP-001 through EXP-012 under `problems/algebraic-geometry/jacobi` each experiment declares its hypothesis before the run and records an adversarial-validation route with its verdict.

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